TERRAIN CLASSIFICATION FOR SMALL-LEGGED ROBOTS

Using Deep Learning on Tactile Data | Machine Learning Mini-Project | Team B-29

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Project summary. We benchmarked three classical classifiers against a CNN for eight-class terrain recognition using 5,990 tactile-data samples. Random Forest achieved the strongest held-out test performance (77.09% accuracy; 0.7698 macro F1), demonstrating the value of evaluating classical baselines alongside deep learning.

1. Problem Statement and Objective

Legged robots encounter surfaces with different contact characteristics. Recognizing terrain from tactile measurements can provide useful information for locomotion systems. This project formulates terrain recognition as an eight-class classification task: asphalt, cork, grass, gravel, lab, laminate wood, pebble and sand. The objective is to compare model families under a common data split and determine which performs best on the available dataset.

2. Dataset and Data Integrity

Dataset property	Details	Dataset property	Details
Samples	5,990	**Input dimensions**	447 features/sample
Tactile signal	414 readings = 6 channels × 69 positions	**Engineered features**	33 per sample
Classes	8 terrain categories	**Split**	4,193 / 898 / 899 (70/15/15%)

The notebook verifies the expected 5,990 × 447 feature shape, checks that labels in the feature and label files agree, and asserts that feature values contain no missing or non-finite values. Stratified splitting preserves class proportions as closely as possible (seed = 42). The original dataset source is not documented in the supplied repository; its provenance should be confirmed separately if a formal dataset citation is required.

3. Preprocessing and Experimental Protocol

The training set is used to fit models, the validation set supports model selection and CNN early stopping, and the held-out test set is reserved for final reporting. Logistic Regression and RBF-SVM use StandardScaler pipelines fitted on training data; Random Forest uses the original feature scales. For the CNN, the 414 tactile readings are reshaped to 6 × 1 × 69 and separated from the 33 engineered features. Both CNN input groups are standardized using training-set statistics only, then the same statistics are applied to validation and test data. This prevents preprocessing from using statistics computed from the held-out sets.

4. Models and Implementation

Model	Implementation	Rationale
Logistic Regression	447 standardized features; max_iter = 2,000	Linear baseline for comparison
SVM (RBF)	447 standardized features; C = 10; gamma = scale	Nonlinear decision boundaries
Random Forest	400 trees; min_samples_leaf = 2; balanced class weights	Tree ensemble for nonlinear feature interactions
CNN	4 Conv2d layers; 24 filters each; (1×5) kernels; max-pooling; dropout 0.5; merge with 33 features; Dense 50; 8 logits	Learns a representation from tactile readings and combines engineered features

CNN training uses CrossEntropyLoss and Adam (learning rate 0.0001), with batch size 64 and a maximum of 100 epochs. Early stopping uses validation accuracy with patience 12, and the best validation checkpoint is restored before testing. The CNN output consists of eight class logits; the predicted class is the highest-scoring output.

Experimental workflow

DATA CHECKS 5,990 × 447	STRATIFIED SPLIT 70 / 15 / 15	FIT + VALIDATE 4 models	HELD-OUT TEST Accuracy + macro F1

5. Results on the Held-Out Test Set

Model	Accuracy	Macro F1	Interpretation
Random Forest	**77.09%**	**0.7698**	Best overall result in this experiment
SVM (RBF)	70.41%	0.7093	Second-best result
CNN	61.96%	0.5900	Class-specific weaknesses remain
Logistic Regression	56.06%	0.5593	Lowest among tested models

Table 1. Final metrics on the 899-sample held-out test set from one stratified split.

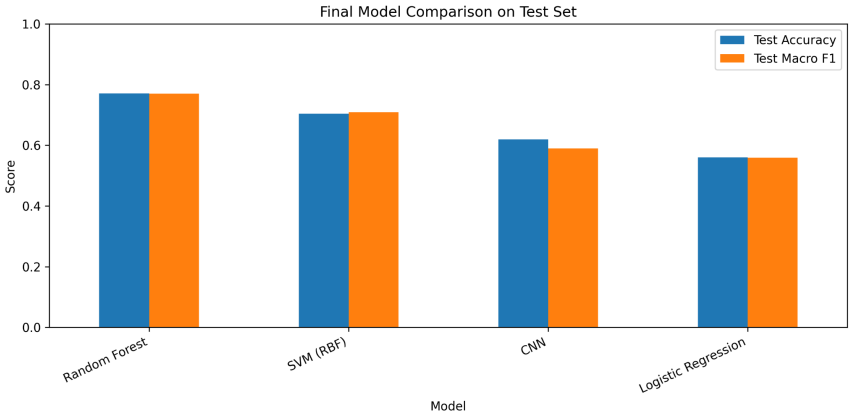


Figure 1. Test accuracy and macro F1 for all four models.

Metric interpretation. Accuracy is the proportion of correct predictions. Macro F1 is the unweighted average of the eight class-wise F1-scores, so each terrain class contributes equally to this metric.

Main finding. Random Forest exceeded CNN test accuracy by 15.13 percentage points and also achieved a higher macro F1. A plausible explanation is that the tree ensemble effectively uses nonlinear relationships in the tabular feature set. This experiment does not isolate the cause of the performance gap; feature ablation and repeated splits would be needed to test that explanation. The result is specific to this dataset, split and model configuration.

6. CNN Class-Level Error Analysis

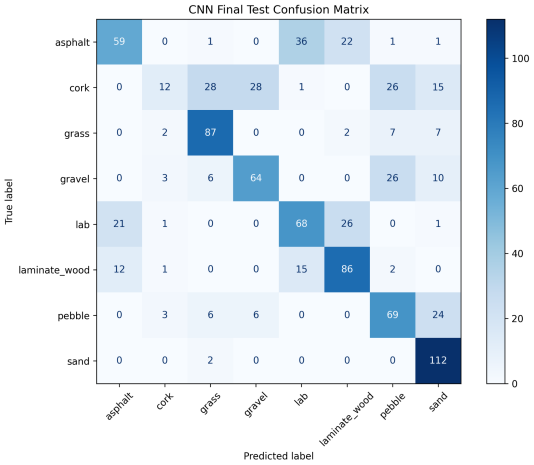


Figure 2. CNN confusion matrix: rows are true labels and columns are predicted labels.

CNN recall was strongest for sand (98%) and grass (83%), followed by laminate wood (74%), pebble (64%), gravel (59%), lab (58%) and asphalt (49%). Cork recall was only 11%: 12 of 110 actual cork samples were correctly classified. Cork was often predicted as grass, gravel or pebble. The confusion matrix identifies these errors but does not by itself establish their physical cause.

7. Limitations, Future Work and Conclusion

This evaluation uses one stratified split and one training run; results may vary across splits or random seeds. Cross-validation, feature importance, feature-group ablation and real-robot deployment have not yet been evaluated. Next steps are to compare tactile-only, engineered-feature-only and combined inputs; repeat evaluation with stratified cross-validation; inspect class-specific errors; and test robustness on data from different sensors or environments. In the current experiment, Random Forest is the strongest of the four evaluated models (77.09% accuracy; 0.7698 macro F1). The findings support systematic comparison of classical and deep-learning methods rather than assuming a CNN will perform best.

Reproducibility: The accompanying notebook documents data checks, split construction, preprocessing, model fitting, validation-based CNN checkpoint selection, test evaluation and saved result artifacts. The dataset's source remains to be verified.